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# Exploring the History of Southeast Asian Astronomy

A Review of Current Projects and Future  
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# Chapter 18

## Star Patterns in Mandar Navigation



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### 18.1 Introduction

#### 18.1.1 Trade and a Seafaring Tradition

Indonesia is a country that consists of thousands islands. In ancient times, the main link between islands was waterborne transport. In addition, Indonesia's coastline is very long, at around 81,000 km. Therefore, maritime activities were quite important in Indonesian society (e.g. see Lapain, 1996; Lampe, 2016).

In the course of Indonesian history, maritime civilizations influenced governments, defense and security, sea transportation and trading, the management of ports, and the law of the sea. These characterized the main maritime kingdoms of the archipelago, such as Srivijaya, Tarumanegara, Majapahit, Banten, Samudera Pasai Aceh, Gowa, Makassar, and the two sultanates of Buton and Ternate. This maritime tradition was mostly carried out by ethnic groups such as the Bugis Makassar, Mandar, Bajo, Buton and Madura, and some Javanese communities. As a matter of fact, this seafaring tradition was associated with Indonesia's important location in Southeast Asia, as it lay on three important international trade routes: the 'silk route' that connected China and Constantinople; the 'ceramic route' to China; and the 'spice route' that connected Indonesia to Europe and the Mediterranean Sea (Tan, 2010).

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### ***18.1.2 The Ethnic Groups of Sulawesi***

Ethnic groups are related social groups based on ancestry and place of origin. Ethnic identity was ascribed at birth, and members of an ethnic group customarily acquired their characteristics during childhood. Later on, these characteristics became the main supports during life, because they reflected identity and honor. As mentioned previously, some ethnic groups in Sulawesi that were (and still are) famous for their distinctive cultural characteristics are the Bugis Makassar, the Mandar and the Bajo. They are famous as traders and as seafarers (Lampe, 2016). Their places of origin are shown in Figure 18.1.

### ***18.1.3 Navigation Using Star Patterns***

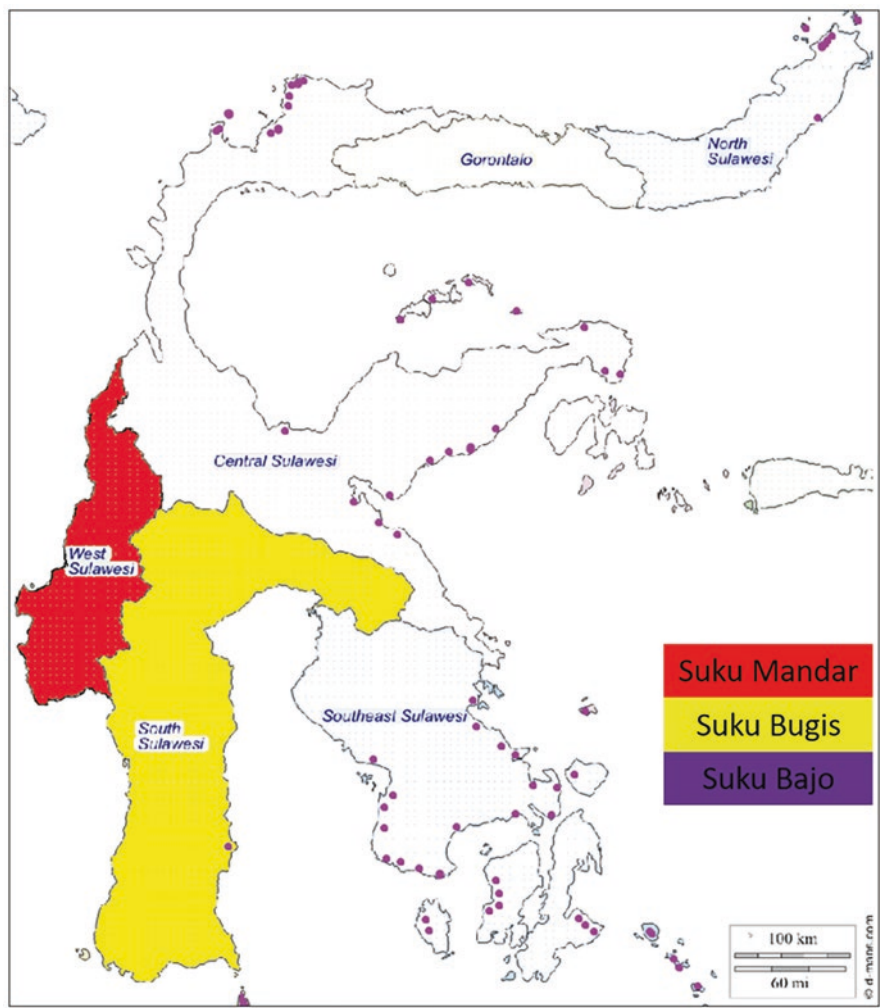
The sailing tradition that has developed since ancient times makes the sea the life-blood of those ethnic groups, especially those inhabiting the coast. They continuously migrate to or trade with other islands, and capture and utilize marine resources, not just for their own consumption but also to be traded. Navigating the seas of the archipelago often required fishermen or traders to spend weeks, or even months, away from home. In order not to get lost during these long voyagers, the sailors had to know how to navigate.

For these voyages around the Archipelago, and even as far afield as the northern coasts of Australia, the seafarers used wooden ships equipped with sails (e.g. see Fig. 18.2). Until recently, they did not have access to modern navigation equipment such as magnetic compasses, binoculars, or sextants, and of course the GPS. Instead, out in the open ocean with no land in sight, the sailors had to use celestial bodies for navigation. During the day, undoubtedly they used the Sun as a benchmark, and also the Moon if it happened to be visible during the day. But at night they recognized certain stars and groups of stars (asterisms) that allowed them to pinpoint the cardinal directions. And so they developed a body of knowledge involving key stars and star patterns, that traditional navigators passed down from generation to generation.

## **18.2 The Mandar Ethnic Group**

### ***18.2.1 Short History of the Mandar Ethnic Group***

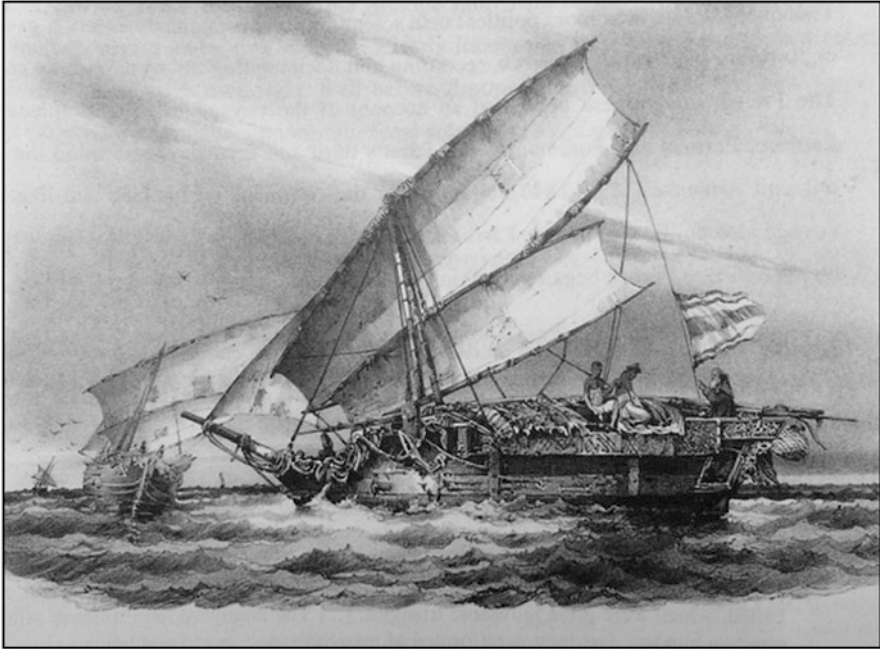
The Mandar ethnic group is centred on the West Sulawesi area (see Fig. 18.1). Presently, West Sulawesi has the status of a new province in Indonesia, created in 2006, after being split from the province of South Sulawesi. The location of the Mandar region stretches from South to North, namely from Binanga Karaeng in the South to Suromana in the North, with the following limits:



**Fig. 18.1** Places of origin of Bugis, Mandar, and Bajo ethnic groups in Sulawesi (map: A. Rasyid).

- In the south it is bordered by Pinrang Regency and Tanah Toraja Regency
- In the north it is bordered by Tanah Toraja and Luwu Regencies
- In the north bordering Central Sulawesi Province
- In the west it is bordered by the Makassar Strait.

The history of the Mandar ethnic group was first recorded in the the sixteenth century (Mandra, 1986). The Mandar area used to consist of several different kingdoms. Originally there were seven kingdoms, and they united into one constitutional organization to form a federation called '*Pitu Babana Binanga*'. The seven original kingdoms were:



**Fig. 18.2** For hundreds of years fleets of *prahu* like these, drawn in 1839, made the long and navigationally-challenging trip from Sulawesi to the northern coast of Australia in order to collect the sea cucumber, a Chinese delicacy (<https://www.abc.net.au/news/2018-01-16/aboriginal-people-asians-trade-before-european-settlement-darwin/9320452>).

- The Kingdom of Balanipa
- The Tjenrana Kingdom
- The Kingdom of Banggae
- The Kingdom of Pambaoeang
- The Tappalang Kingdom
- The Mamuju Kingdom, and
- The Kingdom of Binuang.

The literal meaning of '*Pitu Babana Binanga*' is the 'Seven Estuaries of the River'. This naming is based on the geographical location of the kingdoms and their proximity to the coast.

In addition, for the sake of strategy, both for war and trade, '*Pitu Babana Binanga*' re-established alliances with seven other kingdoms located inland, in the mountains:

- The Kingdom of Rattebulahan
- The Kingdom of Aralle
- The Mambi Kingdom
- The Bambang's Kingdom
- The Kingdom of Messawa

- The Tabulahan Kingdom, and
- The Matangnga Kingdom.

The name of these seven kingdoms federation is ‘*Pitu Ulunna Salo*’, which means ‘Seven Upper Section of the River’. The name is in accordance with the location of the seven kingdoms, which are located in an inland mountainous area where there is a tributary of the river.

### ***18.2.2 The Character of the Mandar Ethnic Group***

Given the geographical location of the Mandar ethnic group, it is no surprise that in the past many Mandar people were fishermen or merchants. While fishing was mainly carried out in the oceans around Sulawesi, as we have seen from Fig. 18.2 Mandar fisherman-merchants sometimes ventured as far afield as the northern coasts of Australia for sea cucumbers (*trepang*).

Like their Bugis neighbours (see Ammarell, 1999), the Mandar acquired an international reputation as merchants, throughout Indonesia. Apart from selling their marine catches they traded spices and merchandise. In the West Sulawesi region, and especially the Majene District, the Mandar produced woven fabrics, silk sarongs, copra, rattan, wood, coconut oil and gold powder. These were processed into materials used by the Mandar community, especially during the west monsoon season when fishermen could not go to sea due to unfavorable weather. Meanwhile, the weaving was performed by women, especially the wives of sailors who were away on lengthy voyages, and they used their spare time while awaiting the return of their husbands.

### ***18.2.3 Trade and Agriculture***

Traditional Mandar voyages are widely known, and went in all four cardinal directions:

- Eastern voyages went to Maluku, Ternate, Kei Island, Aru, Tanimbar, Papua and North Australia. The main aim of these voyages was to buy spices and copra, and catch sea cucumbers.
- Western voyages went to Sumatra, Kalimantan (Borneo), Java, Singapore, Malacca, Cambodia and Siam (Thailand). The main purpose of these voyages was to sell proceeds from the east and bring goods obtained during the voyage back to Sulawesi to be resold.
- Northern voyages went to ports in North Sulawesi and the Philippines.
- Southern voyages went to Java, and even further afield to the islands of West Nusa Tenggara and East Nusa Tenggara, including East Timor.

To facilitate these voyages, the Mandar took advantage of the monsoons: during June–September they went in a westerly direction with the southeast monsoon, and returned to Sulawesi between December and March with the northwest monsoon.

With all of this voyaging, some Mandar people chose to settle in areas that they visited. In his dissertation, Lopa (1982) mentions that apart from the western coast of Sulawesi, Mandar people are also found in

- Almost all of the islands in the Makassar Strait and the Flores Sea;
- Almost all of islands to the south, such as Pulau Marabatua, Pulau Denawang, Pulau Payongpayongan, and those in the Pulau Sembilan District;
- Balikpapan, Samarinda; and
- Java, Nusa Tenggara, Sumatra and in Sandakan, East Malaysia.

## 18.3 Mandar Navigation

### 18.3.1 Research Methodology

The research methodology used in this study was through interviews with retired fishermen, and the first two authors followed up with field observations for verification. All of the fishermen were more than 70 years old, and were known to have knowledge about Mandar star patterns. Unfortunately, we did not have an opportunity to go sailing with them. We noted, however, that the younger fishermen did not use star patterns for their navigation.

Next, we conducted an astronomical analysis by identifying the stars that formed the asterisms in question and determining when they would appear. We used *Stellarium* astronomical software to simulate the star maps and to identify the star patterns. The date of the simulation was set at around the beginning of the twentieth century (January 1900) and the location of the observer was set at the city of Majene in West Sulawesi.

### 18.3.2 Mandar Star Patterns

Star constellations are formed from several stars. Mandar ethnic groups used these star constellations to navigate during night-time sailing. We note, however, that not all direction-indicators were stars or star patterns because celestial objects also included Venus and a star cluster.

Star patterns or asterisms that we identified during this study were *Bittoéng Sapo Keping*, *Bittoéng Tuwallu*, *Bittoéng Mangiwang*, *Bittoéng Lambaru*, *Bittoéng Panjala*, *Bittoéng Malunus*, *Bittoéng Tallu-tallu*, *Bittoéng Pambawa Allo*, *Bittoéng Bawi*, and *Bittoéng Naga*. In astronomical terms, they comprise seven star patterns, the Milky Way, the Pleiades star cluster, and the planet Venus, all of which were

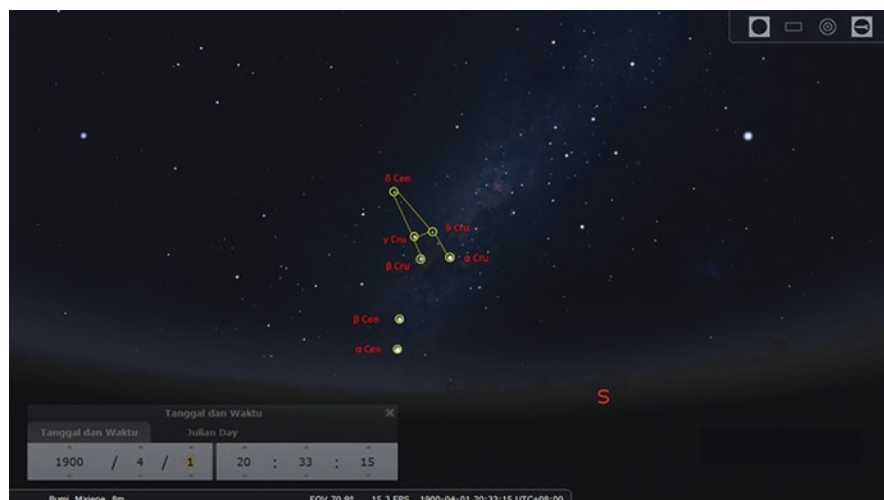
used for navigation. This shows that the practice of astronomical navigation has long been used by Mandar seafarers.

*Bittoéng*, above, means ‘star’, or *Bintang* in the Indonesian language and *Bintoéng* in the Bugis language. The following is an explanation for each star or star pattern.

(1) *Bittoéng Sapo Kepang*: This means a slanted house. This star pattern is used as an indicator for the south (see Fig. 18.3). Usually one can begin to observe this star pattern in the evening sky in May. We can see five stars that form a house with uneven ‘foundations’. The part that forms *Bittoéng Sapo Kepang* is the roof and two supporting pillars, as shown in Fig. 18.3. Comparison with star maps shows that the roof consists of stars  $\delta$  Cen,  $\gamma$  Cru and  $\delta$  Cru, where  $\delta$  Cen is the roof top. The two pillars are formed by  $\beta$  Cru and  $\gamma$  Cru (the short pillar), and  $\alpha$  Cru and  $\delta$  Cru (the long pillar). The stars that form this asterism are listed in Table 18.1.

(2) *Bittoéng Tuwallu*: This means the widow star. This ‘star’ is actually a pair of stars just below the *Bittoéng Sapo Kepang*, so it is also a sign for the south direction. Usually one can begin to observe these stars in the evening of May. We see two stars that are bright and close together and have colors that look a little different: one red, the other green. We know that the ‘red widow’ is  $\alpha$  Centauri and the ‘green widow’ is  $\beta$  Centauri. These two stars are also listed in Table 18.1 and shown in Fig. 18.3.

(3) *Bittoéng Mangiwang*: This means the shark stars. This asterism marks the entry of the northwest monsoon season. It usually rises in the morning at the end of December. You will see a triangular shark fin shape in the middle and a tail at the back. The parts that make up the shape of this shark are the fins, head, stomach and tail, as shown in Fig. 18.4. Our identification shows that the fin is formed by



**Fig. 18.3** The shape of *Bittoéng Sapo Kepang* and its constituent stars seen from the location of the city of Majene (simulated using Stellarium). *Bittoéng Tuwallu*, identified as  $\alpha$  Cen and  $\beta$  Cen, is also shown.

**Table 18.1** List of the stars forming *Bittoeng Sapo Keping* and *Bittoeng Tuwallu*.

| Star Pattern                | Star         | RA                                                 | DEC            |
|-----------------------------|--------------|----------------------------------------------------|----------------|
| <i>Bittoéng Sapo Keping</i> | $\alpha$ Cru | 12 <sup>h</sup> 26 <sup>m</sup> 36.28 <sup>s</sup> | −63° 06′ 00.4″ |
|                             | $\beta$ Cru  | 12 <sup>h</sup> 47 <sup>m</sup> 43.99 <sup>s</sup> | −59° 41′ 20.1″ |
|                             | $\gamma$ Cru | 12 <sup>h</sup> 31 <sup>m</sup> 11.59 <sup>s</sup> | −57° 06′ 21.3″ |
|                             | $\delta$ Cru | 12 <sup>h</sup> 15 <sup>m</sup> 09.26 <sup>s</sup> | −58° 44′ 56.9″ |
|                             | $\delta$ Cen | 12 <sup>h</sup> 08 <sup>m</sup> 21.88 <sup>s</sup> | −50° 43′ 19.5″ |
| <i>Bittoéng tuwallu</i>     | $\alpha$ Cen | 14 <sup>h</sup> 40 <sup>m</sup> 09.60 <sup>s</sup> | −60° 55′ 19.1″ |
|                             | $\beta$ Cen  | 14 <sup>h</sup> 03 <sup>m</sup> 49.99 <sup>s</sup> | −60° 22′ 24.5″ |



**Fig. 18.4** The shape of *Bittoéng Mangiwang* and its constituent stars seen from the location of the city of Majene (simulated using Stellarium).

stars  $\zeta$  Sco,  $\lambda$  Sco,  $\upsilon$  Sco,  $\kappa$  Sco and  $\iota 1$  Sco, where  $\upsilon$  Sco is the upper end of the fin. The head part consists of  $\alpha$  Ara,  $\beta$  Ara and  $\theta$  Ara, where  $\beta$  Ara is the tip of the head. The abdomen consists of  $\alpha$  Tel and  $\epsilon$  Sgr. The tail part of the shark is formed from stars  $\delta$  Sgr,  $\mu$  Sgr,  $\lambda$  Sgr,  $\phi$  Sgr and  $\sigma$  Sgr, where  $\delta$  Sgr connects the tail and the body of the shark. The stars that form this asterism are listed in Table 18.2.

(4) *Bittoéng Lambaru*: This means stingray stars. This asterism is an early clue to the arrival of the northwest season and usually rises in the morning at the end of November. There are two eyes in the shape of a stingray’s head that are above *Bittoéng Mangiwang*. The parts that make up the stingray’s shape are the head and tail, as shown in Fig. 18.5. The head of the stingray is formed by the stars  $\beta 1$  Sco,  $\delta$  Sco,  $\eta$  Sco,  $\rho$  Sco and  $\sigma$  Sco, where  $\delta$  Sco and  $\eta$  Sco are the stingray’s eyes and  $\sigma$  Sco is a connector between the tail and the head. In addition, the stingray’s tail consists of the stars  $\alpha$  Sco,  $\tau$  Sco,  $\epsilon$  Sco,  $\mu 1$  Sco and  $\zeta 2$  Sco. The stars that form this asterism are also listed in Table 18.3.

**Table 18.2** List of stars forming *Bittoéng Mangiwang*.

| Star Pattern              | Star   | RA                                                 | DEC            |
|---------------------------|--------|----------------------------------------------------|----------------|
| <i>Bittoéng Mangiwang</i> | α Ara  | 17 <sup>h</sup> 31 <sup>m</sup> 50.89 <sup>s</sup> | −49° 52′ 27.6″ |
|                           | β Ara  | 17 <sup>h</sup> 25 <sup>m</sup> 18.31 <sup>s</sup> | −55° 31′ 45.2″ |
|                           | θ Ara  | 18 <sup>h</sup> 06 <sup>m</sup> 37.94 <sup>s</sup> | −50° 05′ 28″   |
|                           | α Tel  | 18 <sup>h</sup> 19 <sup>m</sup> 46.38 <sup>s</sup> | −45° 58′ 00.3″ |
|                           | γ Sgr  | 18 <sup>h</sup> 05 <sup>m</sup> 49.23 <sup>s</sup> | −30° 25′ 06.0″ |
|                           | δ Sgr  | 18 <sup>h</sup> 20 <sup>m</sup> 59.44 <sup>s</sup> | −29° 49′ 38.2″ |
|                           | ε Sgr  | 18 <sup>h</sup> 24 <sup>m</sup> 11.07 <sup>s</sup> | −34° 22′ 55.2″ |
|                           | λ Sgr  | 18 <sup>h</sup> 27 <sup>m</sup> 58.74 <sup>s</sup> | −25° 24′ 58.1″ |
|                           | μ Sgr  | 18 <sup>h</sup> 13 <sup>m</sup> 58.81 <sup>s</sup> | −21° 03′ 31.7″ |
|                           | σ Sgr  | 18 <sup>h</sup> 55 <sup>m</sup> 15.84 <sup>s</sup> | −26° 17′ 42.4″ |
|                           | φ Sgr  | 18 <sup>h</sup> 45 <sup>m</sup> 39.02 <sup>s</sup> | −26° 59′ 26.9″ |
|                           | θ Sco  | 17 <sup>h</sup> 37 <sup>m</sup> 19.08 <sup>s</sup> | −42° 59′ 52.0″ |
|                           | ι1 Sco | 17 <sup>h</sup> 40 <sup>m</sup> 35.09 <sup>s</sup> | −40° 07′ 36.5″ |
|                           | κ Sco  | 17 <sup>h</sup> 42 <sup>m</sup> 29.34 <sup>s</sup> | −39° 01′ 45.1″ |
|                           | λ Sco  | 17 <sup>h</sup> 33 <sup>m</sup> 36.49 <sup>s</sup> | −37° 06′ 10.7″ |
|                           | ς Sco  | 17 <sup>h</sup> 49 <sup>m</sup> 51.11 <sup>s</sup> | −37° 02′ 39.0″ |
|                           | υ Sco  | 17 <sup>h</sup> 30 <sup>m</sup> 45.00 <sup>s</sup> | −37° 17′ 42.1″ |

(6) *Bittoéng Malunus*. This term has no special meaning, but marks the Pleiades star cluster. This asterism (Fig. 18.6) is used as a clue to the increasing number of flying fish (or fish *tipping-tuing*) in the Makassar Strait during the southeast monsoon. It usually rises in the east at 5 o’clock in the morning at the end of June.

(7) *Bittoéng Bawi*. This means a pig star. This star is usually used by mountain people as a cautious sign because pigs will come out looking for food. This is actually the planet Venus.

(8) *Bittoéng Pambosei*. This means the star of fishermen. This star is a marker towards the north. This star usually appears in February, part way through the north-west monsoon.

(9) *Bittoéng Naga*. This means the ‘dragon star’. This ‘star’ is used by fishermen for a good fishing location. It usually rises in the southeast. When this ‘star’ disappears, fishermen will be afraid because strong wind will show up. This is actually not a ‘star’ at all, but is the Milk Way.

(10) *Bittoéng Panjala*. This means the fishing net stars. This asterism is used as an indicator of the entry of the cool season (between the two monsoons) and the emergence of the bambang fish. This asterism usually rises around midnight in April. The parts that make up *Bittoéng Panjala* are the fishermen and the boat, as shown in Fig. 18.7. The identification shows that the boat is formed from the stars α UMa, β UMa, λ Dra, κ Dra, α Dra and θ Boo. In addition, fishermen are formed by stars γ UMa, δ UMa, ε UMa, ζ UMa and η UMa, where δ UMa, ε UMa and ζ UMa are the fishermen and γ UMa and η UMa are fishing equipment. The stars that form this asterism are also listed in Table 18.4.

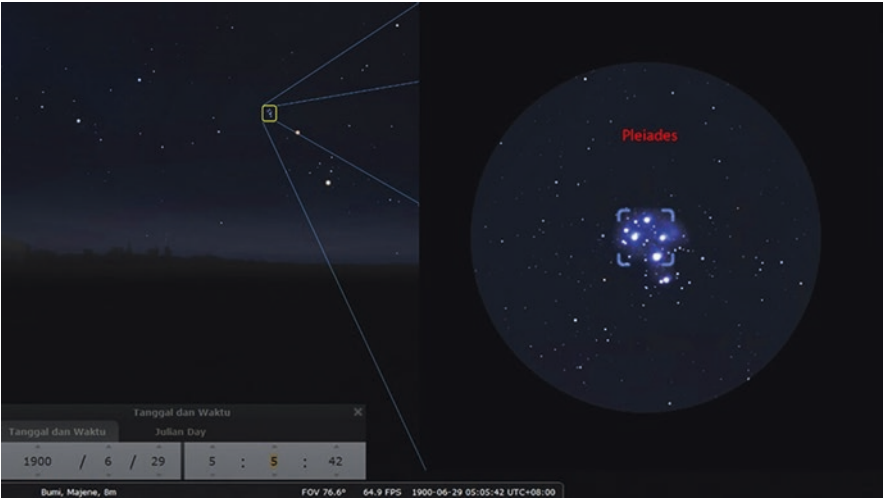


**Fig. 18.5** The shape of *Bittoéng Lambaru* and its constituent stars seen from the location of the city of Majene (simulated using Stellarium).

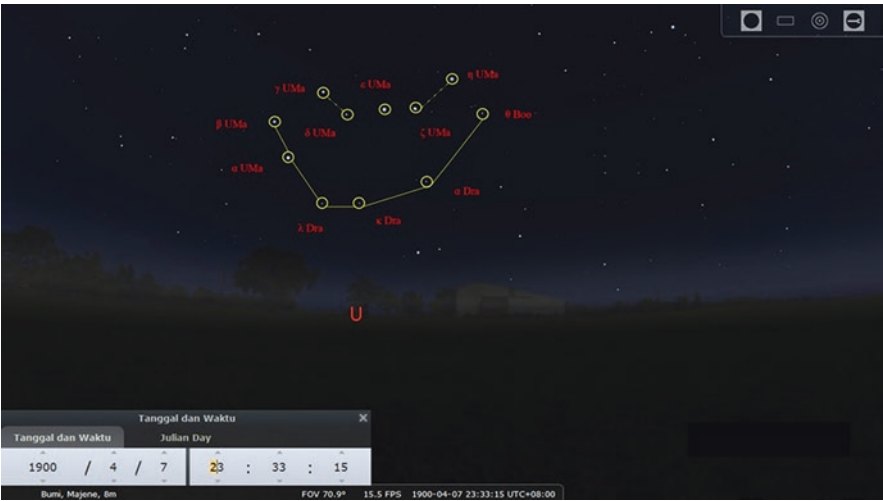
**Table 18.3** List of stars forming *Bittoéng Lambaru*.

|                         |                |                                                    |                 |
|-------------------------|----------------|----------------------------------------------------|-----------------|
| <i>Bittoéng Lambaru</i> | $\alpha$ Sco   | 16 <sup>h</sup> 29 <sup>m</sup> 24.53 <sup>s</sup> | -26° 25' 52.7'' |
|                         | $\beta$ 1 Sco  | 16 <sup>h</sup> 00 <sup>m</sup> 26.27 <sup>s</sup> | -19° 48' 16.8'' |
|                         | $\delta$ Sco   | 16 <sup>h</sup> 00 <sup>m</sup> 20.06 <sup>s</sup> | -22° 37' 13.8'' |
|                         | $\epsilon$ Sco | 16 <sup>h</sup> 50 <sup>m</sup> 14.27 <sup>s</sup> | -34° 16' 54.5'' |
|                         | $\eta$ Sco     | 15 <sup>h</sup> 58 <sup>m</sup> 51.9 <sup>s</sup>  | -26° 06' 47.8'' |
|                         | $\mu$ 1 Sco    | 16 <sup>h</sup> 51 <sup>m</sup> 52.28 <sup>s</sup> | -38° 02' 48.0'' |
|                         | $\zeta$ 2 Sco  | 16 <sup>h</sup> 54 <sup>m</sup> 35.43 <sup>s</sup> | -42° 21' 16.0'' |
|                         | $\rho$ Sco     | 15 <sup>h</sup> 56 <sup>m</sup> 53.18 <sup>s</sup> | -29° 12' 48.0'' |
|                         | $\sigma$ Sco   | 16 <sup>h</sup> 21 <sup>m</sup> 11.38 <sup>s</sup> | -25° 35' 32.1'' |
|                         | $\tau$ Sco     | 16 <sup>h</sup> 35 <sup>m</sup> 53.00 <sup>s</sup> | -28° 12' 55.4'' |

- (11) *Bittoéng Tallu-tallu*. This means a group of three stars. These stars are a sign towards the east after *Bittoéng Malunus*. One may observe three stars in a line which are quite bright at dawn. The parts that make up *Bittoéng Tallu-tallu* are two groups of stars that each have three stars in line, as shown in Fig. 18.8. Obviously these are in the modern constellation of Orion. The first part consists of stars  $\delta$  Ori,  $\epsilon$  Ori and  $\zeta$  Ori. The other part consists of  $\pi$ 3 Ori,  $\pi$ 4 Ori, and  $\pi$ 5 Ori. The stars that form this asterism are listed in Table 18.5.
- (12) *Bittoéng Pambawa Allo*. This means a star carrying the Sun. This star is a sign that the Sun will rise about an hour later. It is usually seen around 4 o'clock in the east. Actually this is the planet Venus.



**Fig. 18.6** The form of *Bittoeng Malunus* seen from the location of the city of Majene (simulated using Stellarium).



**Fig. 18.7** The shape of *Bittoeng Panjala* and its constituent stars seen from the location of the city of Majene (simulated using Stellarium).

## 18.4 Concluding Remarks

We have identified eleven celestial patterns in Mandar navigation, but only six of them related to star patterns useful for navigation. The Mandar view of star patterns is obviously related to the livelihoods of the fishermen, such as the patterns

of fish, a boat and a fishing net. Other celestial patterns are used as a sign of the presence of fish in significant numbers or particular environmental features. Presently, many young Mandar sailors and fishermen have little knowledge of stars and other environmental features and rely exclusively on the magnetic compass and charts to guide their ships. So it is very important to retain the old navigational methods that use celestial bodies, as this is part of the rich traditional culture of the Mandar people.

In Table 18.6, we compare the star names of the Mandar and Bugis, two ethnic groups that have a strong seafaring tradition. We see that the asterisms are similar, both in name and in form, which shows the close cultural links between these two neighbouring ethnic groups.

We should not be surprised by these similarities, for the Bugis and the Mandar both owe their oceanic maritime prowess and their astronomical systems to the Austronesians, who migrated from Taiwan to the Philippines about 4000 years ago. One branch of the population then moved eastwards, and eventually out into the Pacific (Carson et al., 2013), while the other branch moved southwards into Sulawesi, bringing with them (amongst other things) rice cultivation, domesticated pigs and a distinctive red-slipped pottery (Bellwood, 2017). Evidence of these early Austronesian settlers has been excavated at a number of archaeological sites in the Karama Valley, where radio carbon dates indicates their arrival around 3,500 years ago (e.g. see Anggraeni et al., 2014). There also is strong genetic evidence that supports this Austronesian settlement of Sulawesi (see Lipson et al., 2014). Non-coastal groups and some coastal people were committed to agriculture, but other coastal groups decided to focus instead on fishing and maritime commerce, honing the highly advanced navigational skills and oceanic technology introduced by their Austronesian forefathers.

Elsewhere in this book, Khairuddin and Jaafar (2021) have proposed that among Austronesian speakers in Southeast Asia there is a correlation between asterisms that an ethnic group recognizes in the sky and their ecological links to the local

**Table 18.4** List of star-forming stars in *Bittoéng Panjala*.

| Star Pattern            | Star  | RA                                                 | DEC           |
|-------------------------|-------|----------------------------------------------------|---------------|
| <i>Bittoéng Panjala</i> | α UMa | 11 <sup>h</sup> 03 <sup>m</sup> 44.98 <sup>s</sup> | 61° 44′ 50.4″ |
|                         | β UMa | 11 <sup>h</sup> 01 <sup>m</sup> 49.31 <sup>s</sup> | 56° 22′ 56.2″ |
|                         | λ Dra | 11 <sup>h</sup> 31 <sup>m</sup> 24.80 <sup>s</sup> | 69° 19′ 47.9″ |
|                         | κ Dra | 12 <sup>h</sup> 33 <sup>m</sup> 29.57 <sup>s</sup> | 69° 47′ 23.4″ |
|                         | α Dra | 14 <sup>h</sup> 04 <sup>m</sup> 24.20 <sup>s</sup> | 64° 22′ 36.8″ |
|                         | θ Boo | 14 <sup>h</sup> 25 <sup>m</sup> 13.79 <sup>s</sup> | 51° 51′ 45.7″ |
|                         | γ UMa | 11 <sup>h</sup> 53 <sup>m</sup> 48.69 <sup>s</sup> | 53° 41′ 39.8″ |
|                         | δ UMa | 12 <sup>h</sup> 15 <sup>m</sup> 24.26 <sup>s</sup> | 57° 01′ 51.9″ |
|                         | ε UMa | 12 <sup>h</sup> 54 <sup>m</sup> 0.38 <sup>s</sup>  | 56° 30′ 08.6″ |
|                         | ζ UMa | 13 <sup>h</sup> 23 <sup>m</sup> 54.05 <sup>s</sup> | 54° 55′ 30.4″ |
|                         | η UMa | 13 <sup>h</sup> 47 <sup>m</sup> 33.65 <sup>s</sup> | 49° 18′ 51.6″ |



**Fig. 18.8** The shape of *Bittoéng Tallu-tallu* and its constituent stars seen from the location of the city of Majene (simulated using Stellarium).

**Table 18.5** List of stars forming *Bittoéng Tallu-tallu*.

| Star Pattern                | Star   | RA                                                | DEC           |
|-----------------------------|--------|---------------------------------------------------|---------------|
| <i>Bittoéng Tallu-tallu</i> | δ Ori  | 5 <sup>h</sup> 31 <sup>m</sup> 0.39 <sup>s</sup>  | −1° 17′ 56.7″ |
|                             | ε Ori  | 5 <sup>h</sup> 36 <sup>m</sup> 12.81 <sup>s</sup> | −1° 12′ 06.8″ |
|                             | ζ Ori  | 5 <sup>h</sup> 40 <sup>m</sup> 45.51 <sup>s</sup> | −1° 56′ 33.5″ |
|                             | π3 Ori | 4 <sup>h</sup> 49 <sup>m</sup> 47.66 <sup>s</sup> | 6° 57′ 39.6″  |
|                             | π4 Ori | 4 <sup>h</sup> 51 <sup>m</sup> 12.38 <sup>s</sup> | 5° 36′ 18.3″  |
|                             | π5 Ori | 4 <sup>h</sup> 54 <sup>m</sup> 15.10 <sup>s</sup> | 2° 26′ 26.4″  |

environment. We have noted that more than half of the Mandar asterisms mentioned in this chapter relate in some way to fishing or maritime activities. It will now be interesting to survey the astronomical knowledge-base of inland Mandar groups engaged in rice cultivation and see if they also recognize the marine asterisms of their ancestors, or whether they have already developed their own suite of agriculturally-inspired asterisms. Or will we find an interesting mix of both traditions?

**Table 18.6** A comparison of Bugis (Ammarell, 1999) and Mandar star names.

| Bugis Name                     | Star               | Mandar Name                 | Star            |
|--------------------------------|--------------------|-----------------------------|-----------------|
| <i>Bintoèng Balè Mangiwèng</i> |                    | <i>Bittoéng Mangiwang</i>   | $\alpha$ Ara    |
|                                |                    |                             | $\beta$ Ara     |
|                                |                    |                             | $\theta$ Ara    |
|                                |                    |                             | $\alpha$ Tel    |
|                                |                    |                             | $\gamma$ Sgr    |
|                                |                    |                             | $\delta$ Sgr    |
|                                |                    |                             | $\epsilon$ Sgr  |
|                                |                    |                             | $\lambda$ Sgr   |
|                                |                    |                             | $\mu$ Sgr       |
|                                |                    |                             | $\sigma$ Sgr    |
|                                |                    |                             | $\varphi$ Sgr   |
|                                | $\pi$ Scorpio      |                             | $\theta$ Sco    |
|                                | $\theta$ Scorpio   |                             | $\iota 1$ Sco   |
|                                | $\iota$ Scorpio    |                             | $\kappa$ Sco    |
|                                | $\kappa$ Scorpio   |                             | $\lambda$ Sco   |
|                                | $\lambda$ Scorpio  |                             | $\varsigma$ Sco |
|                                | $\upsilon$ Scorpio |                             | $\upsilon$ Sco  |
| <i>Bintoèng Lambaruè</i>       | $\alpha$ Scorpio   | <i>Bittoéng Lambaru</i>     | $\alpha$ Sco    |
|                                | $\delta$ Scorpio   |                             | $\beta 1$ Sco   |
|                                | $\tau$ Scorpio     |                             | $\delta$ Sco    |
|                                | $\beta$ Scorpio    |                             | $\epsilon$ Sco  |
|                                | $\epsilon$ Scorpio |                             | $\eta$ Sco      |
|                                | $\mu$ Scorpio      |                             | $\mu 1$ Sco     |
|                                | $\zeta$ Scorpio    |                             | $\zeta 2$ Sco   |
|                                | $\eta$ Scorpio     |                             | $\rho$ Sco      |
|                                |                    |                             | $\sigma$ Sco    |
| <i>Bintoèng Bola Kèppang</i>   | $\alpha$ Crux      | <i>Bittoéng Sapo Kepang</i> | $\alpha$ Cru    |
|                                | $\beta$ Crux       |                             | $\beta$ Cru     |
|                                | $\delta$ Crux      |                             | $\gamma$ Cru    |
|                                | $\gamma$ Crux      |                             | $\delta$ Cru    |
|                                | $\mu$ Cen          |                             | $\delta$ Cen    |
|                                |                    |                             |                 |
| <i>Bintoèng Baluè</i>          | $\alpha$ Centauri  | <i>Bittoéng tuwallu</i>     | $\alpha$ Cen    |
|                                | $\beta$ Centauri   |                             | $\beta$ Cen     |
| <i>Bèmbè'è</i>                 | Coal Sack Nebula   |                             |                 |
| <i>Worong porongngè</i>        | Pleiades           | <i>Bittoéng Malunus</i>     | Pleiades        |
| <i>Bintoèng Pitu</i>           |                    |                             |                 |

(continued)

**Table 18.6** (continued)

| Bugis Name                    | Star                  | Mandar Name                  | Star           |
|-------------------------------|-----------------------|------------------------------|----------------|
| <i>Bintoèng balu Mandara'</i> | $\alpha$ Ursa Mayor   | <i>Bittoéng Panjala</i>      | $\alpha$ UMa   |
|                               | $\beta$ Ursa Mayor    |                              | $\beta$ UMa    |
| <i>Bintoèng Kappala'è</i>     |                       |                              | $\lambda$ Dra  |
|                               |                       |                              | $\kappa$ Dra   |
|                               |                       |                              | $\alpha$ Dra   |
|                               |                       |                              | $\theta$ Boo   |
|                               | $\delta$ Ursa Mayor   |                              | $\gamma$ UMa   |
|                               | $\epsilon$ Ursa Mayor |                              | $\delta$ UMa   |
|                               | $\eta$ Ursa Mayor     |                              | $\epsilon$ UMa |
|                               | $\gamma$ Ursa Mayor   |                              | $\zeta$ UMa    |
|                               | $\zeta$ Ursa Mayor    |                              | $\eta$ UMa     |
| <i>Tanra Tèlluè</i>           | $\delta$ Orion        | <i>Bittoéng Tallu-tallu</i>  | $\delta$ Ori   |
|                               | $\epsilon$ Orion      |                              | $\epsilon$ Ori |
|                               | $\zeta$ Orion         |                              | $\zeta$ Ori    |
|                               |                       |                              | $\pi 3$ Ori    |
|                               |                       |                              | $\pi 4$ Ori    |
|                               |                       |                              | $\pi 5$ Ori    |
| <i>Wari wariè</i>             | Venus                 | <i>Bittoéng Pambawa Allo</i> | Venus          |
| <i>Bintoèng Bawi</i>          |                       | <i>Bittoéng Bawi</i>         |                |
| <i>Bintoèng Nagaè</i>         | Milky Way             | <i>Bittoéng Naga</i>         | Milky Way      |
| <i>Tanra Bajoe</i>            | Awan Magellan         |                              |                |
| <i>Bintoèng Timoro'</i>       | Altair                |                              |                |
| <i>Pajjèkoè</i>               | $\beta$ Orion         |                              |                |

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